

Examiner
only

5. The table gives information about two kettles.

Kettle	Power (W)	Time to boil 0.5 litres of water (minutes)	Time to boil 0.5 litres of water (hours)
A	3000	2	0.033
B	2400	2.5	

- (a) (i) Calculate the units used (kWh) by **kettle B** if it is used to boil 0.5 litres of water 6 times a day. [2]

units used = kWh

- (ii) Kettle B is used 6 times a day.
Calculate the cost of using kettle B for **one week**. [2]
Cost of 1 unit = 34 p.

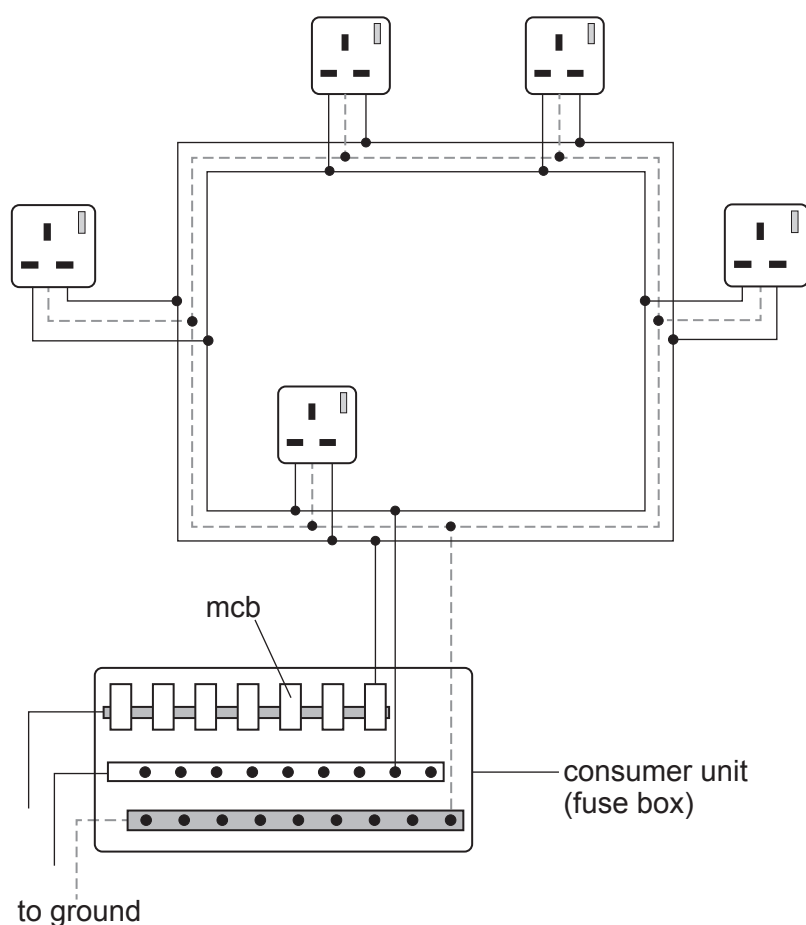
cost = p

- (iii) James suggests that it would be cheaper to use kettle B because it has a lower power.
Sophia disagrees and suggests it would make no difference. [2]
Explain who is correct.
Space for calculation.

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- (b) Kettle B is connected into a ring main circuit.
A diagram of a ring main circuit is shown below.



- (i) State **two** advantages of a ring main circuit. [2]
1.
 2.
- (ii) The ring main is protected by an mcb.
State **two** advantages of an mcb compared to a fuse. [2]
1.
 2.



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- (iii) The mcb has a rating of 32 A.
The maximum power that the ring main can supply is 7360 W.
The table shows some of the appliances that are used in a kitchen.

Appliance	Power (W)
kettle	2400
toaster	1200
microwave	800
washing machine	750
dishwasher	1500
iron	1100

Jack states that all these appliances can be connected to the ring main and used at the same time.
Determine whether Jack is correct.
Space for calculation.

[1]

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